

Vishweshwar Tyagi

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RESEARCH INTERESTS

Bayesian and variational inference, Monte Carlo methods, probability and stochastic processes, analysis and partial differential equations, statistical machine learning

EDUCATION

Columbia University

New York, NY

MS Data Science – GPA 3.97/4

Sep 2021 – Dec 2022

Courses: Machine Learning, Deep Learning, Statistical Inference, Algorithms, Computer Systems

Teaching Assistant: Analysis and Optimization, Applied Machine Learning, Deep Learning, Reinforcement Learning

IIT Kanpur

Kanpur, India

MS Mathematics – GPA 9/10

Aug 2019 – May 2021

Courses: ODE, PDE, Differential Geometry, Topology, Complex Analysis, Probabilistic Programming, Statistical NLP

Honors: Academic Excellence Awards 2019, 2020

University of Delhi

New Delhi, India

BS Mathematics – GPA 9.41/10, Math GPA 9.61/10

Aug 2016 – May 2019

Courses: Multivariate Calculus, Numerical Methods, Series of Functions, Probability and Statistics

RESEARCH EXPERIENCE

Data Scientist

Jan 2023 – Present

Movement Recovery Lab, Department of Neurology, Columbia University

New York, NY

- Developed hierarchical Bayesian model to estimate neural recruitment curves from sparse brain and spinal stimulation data, improving estimation accuracy
- Implemented mixture model to identify and downweight outliers without discarding data, improving robustness
- Developed Bayesian mixed model to detect change in motor threshold, reducing sample size to assess treatment effects in intervention studies
- Developed Bayesian adaptive design to acquire neural recruitment curves using Expected Information Gain, improving acquisition efficiency
- Optimized adaptive design speed in JAX and deployed it using Lab Streaming Layer for online experiments

PUBLICATIONS

Tyagi, V., Murray, L. M., Asan, A. S., Mandigo, C., Virk, M. S., Harel, N. Y., Carmel, J. B., McIntosh, J. R. *Hierarchical Bayesian estimation of motor-evoked potential recruitment curves yields accurate and robust estimates.* Brain Stimulation (2025). <https://doi.org/10.1016/j.brs.2025.09.008>

(In preparation) **Tyagi, V.**, Dukkipati, S. S., Twili, D., Mikhaeil, J. M., Gelman, A., Carmel, J. B., McIntosh, J. R. *Bayesian adaptive design for acquisition of neural recruitment curves.* Expected 2026.

INDUSTRY EXPERIENCE

Data Science Intern

Summer 2022

Quartet Health

New York, NY

- Fine-tuned large language model BERT on clinical notes to identify high-risk patients for mental health conditions, increasing F2-score by 13% over XGBoost baseline
- Built data pipeline on Amazon Redshift using dbt and SQL to automate transformation of medical claims data, added unit tests to validate pipeline output
- Utilized data pipeline to evaluate insurance network quality and identify network gaps, reducing claim denial rates by 7% through outlier detection and saving \$20K in referral costs

SOFTWARE

onMEP Online acquisition of neural recruitment curves

hbMEP Hierarchical Bayesian estimation of neural recruitment curves (<https://hbmep.github.io/hbmep/>)

TEACHING

Teaching Assistant, Columbia University

- Applied deep learning (CS W4995) Fall 2022
- Applied machine learning (CS W4995) Spring 2022
- Analysis and optimization (MATH V2500) Spring 2022
- Reinforcement learning (EE E6885) Fall 2021
- Calculus I (MATH UN1101) Fall 2021

TECHNICAL SKILLS

Programming: Python, R, C/C++, SQL (Postgres)

Probabilistic Programming: NumPyro, Pyro, Stan

Machine Learning: JAX, scikit-learn, XGBoost, Hugging Face, OpenAI

Big Data: dbt, Spark, BigQuery, Redshift

Developer Tools: Git, Bash, Linux, SSH, Vim, VS Code, Docker, Google Cloud Platform

CONFERENCES

Talk: *Hierarchical Bayesian estimation of motor-evoked potential recruitment curves yields accurate and robust estimates.* Society for Neuroscience (SfN) Nanosymposium on Analytical Computational Tools, October 2024, Chicago, Illinois.

Poster: *Hierarchical Bayesian estimation of motor-evoked potential recruitment curves yields accurate and robust estimates.* 11th Annual Minnesota Neuromodulation Symposium – Neuromodulation of Spinal Cord Injury: Translational Opportunities, April 2024, Minneapolis, Minnesota. 3rd Place Poster Award.